

C Language - Loops & Conditions (Short Explanation)

For Loop

Repeats a block of code a fixed number of times.

```
for(int i=0;i<5;i++){  
    printf("%d", i);  
}
```

Initialization → condition check → execution → update. If update is missing, loop can become infinite.

Practice: Print even numbers from 1–20.

While Loop

Executes as long as the condition remains true.

```
int i=1;  
while(i<=5){  
    printf("%d", i);  
    i++;  
}
```

If the condition is false initially, loop will not execute.

Practice: Print numbers 10 to 1.

Do-While Loop

Runs the loop body at least once.

```
int x=1;  
do{  
    printf("%d", x);  
    x++;  
}while(x<=5);
```

Condition is checked after execution, so loop runs at least once.

Practice: Accept input until user enters 0.

If / If-Else

Executes a block based on a condition.

```
if(n%2==0)  
    printf("Even");  
else  
    printf("Odd");
```

If condition is true, first block runs; otherwise, else block executes.

Practice: Check if number is positive, negative, or zero.